

# **VRET**

**Virtual Reality Exposure Therapy - 2026 Fall**

## Returning Members

*(favorite thing on campus)*

1. Jayden Shofolahan (The SAC)

## New Members

*(favorite thing on campus)*

1. Amanda Chen (Jasmine)
2. Coco Gao (SBS)
3. Kevin Chen (Food Trucks)
4. Fahim Jawad (Staller Steps)
5. Carlos Deleon (The waterfall pathway)

# Overall Project's Goal

- Recreate traumatic real-world scenarios in virtual reality that can be used for exposure therapy
- Test the therapeutic tool on actual patients
- Gather qualitative and quantitative data from patients
- Automation of the tools (compatible across many devices)

# Overall timeline

- How does these specific aim fit into a timeline from now to last day of class (Dec 9th)
  - 9/4: Organize Sub-team Goals for Semester and Reach out to experts in psychology and computer science
  - 9/10:
  - 9/18: Work on movement of the characters in scene
  - 9/25: Finish working scene (voice files, characters, everything)
  - 10/16: Work with psychologists to see how to use this scene for trauma patients (experiment)
  - 11/6: Gather test subjects to test the scene on them (n = 10 as an estimate)
  - 11/13-27: Gather test results of the scene for the patients
  - 11/28-12/9: Finalize project and gather all data (all characters, scene working well, qualitative and quantitative data finalized)



# Resources & Documentation

- Google drive:  
<https://drive.google.com/drive/u/0/folders/1Ehx2qt1LO1rx1NGT6xhRWBDbQj5JHleC>
- New members: read the papers attached to the Google Drive and be familiar with VRET Github: <https://github.com/TLTMedia/VRET>
- LAM\_Audio2Expression:  
[https://github.com/aigc3d/LAM\\_Audio2Expression](https://github.com/aigc3d/LAM_Audio2Expression) Valid VRM Avatars:  
<https://github.com/TLTMedia/valid-vrm-avatars>  
Audio2Face<https://build.nvidia.com/nvidia/audio2face-3d>
- Computer Science resources from SBU  
<https://ai.stonybrook.edu/resources/computingresources>

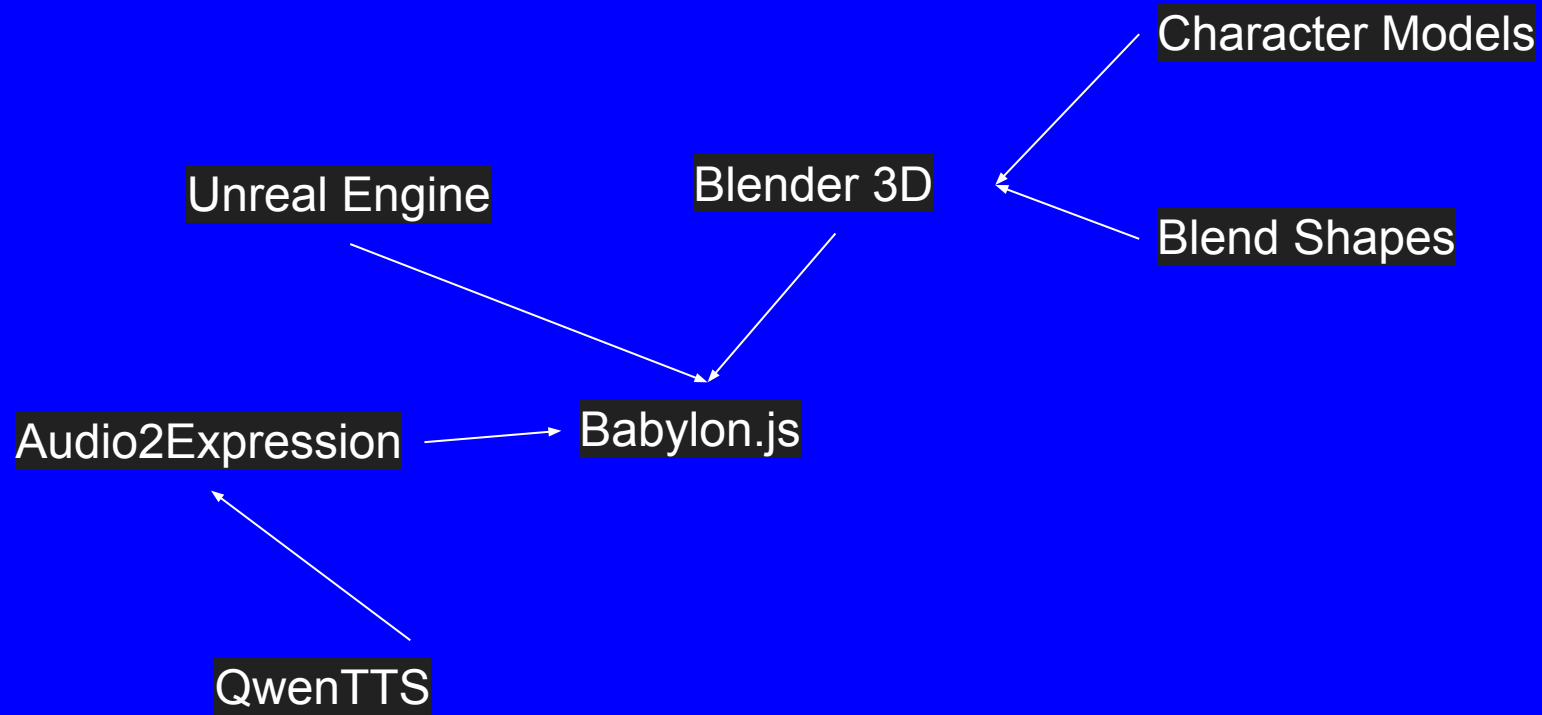
# How to work together

- Timeful (update this: <https://timeful.app/e/9f7A6> )
- Will keep in touch with **Discord and Emails**
- Whole team (with faculty adviser)
  - Meet: **Fri 3:30-4:30PM**
  - Weekly to determine agenda

# Tool Overview

- The project will involve the following technologies/schemes (may be subject to change over time depending on project compatibility issues)
  - Unreal Engine → Foundation for creating levels and scenes
  - Babylon.js → Combines scenes, environment, and animation into a single unit. Tool that we are using to make our environment
  - Blender 3D → Current character build and animation tool.
    - Implements blend shapes for transitional facial animations.
  - ARKit-52 → Substandard within the ARKit Specification. Facial expressions naming scheme for aligning animation with audio.
  - NVIDIA Audio2Face-3D OR LAM Audio2Expression → Determine blend shape weights for facial expression transitions.
  - QwenTTS (Open Source) → Character Voice Lines (offline, generates in the backend)
  - WebXR → Prioritizing this, umbrella for AR and VR, better for compatibility, straightforward to do this in Babylon
  - GitHub → Current file storage system and software build pipeline.
    - Audio, blendshape, character models, scene information combined over [Babylon.js](#).
    - Some will be online tools, some will be offline

# Software Integration Process





# Individual role in a team

Update from first slide submission:

Scene making is done, as well as the script

Divided parts for parallel work across team members

# Jayden's Role

- **Tasks:**

- Refine voice files if needed
- Email psychologists to ask for their input
- Look into the psychiatric and neurological benefits of treating trauma
- Understand the technological implications of using VRET
- Design the scene so that after the scene is finished, the therapy part can begin
- Design novel scenes where it doesn't directly show the violent scene, but rather a purely therapeutic scene that walks the user through their trauma in a way that doesn't make the person traumatized a gain
- Collect data on how test subjects have felt after the treatment (interview questions and the Likert scale, quantitative and qualitative data)

- **Realistic Achievement:**

- Finalized scene and collaboration with psychologists and psychiatrists
- Novel scene where it is strictly therapeutic, for those who don't want a direct re-exposure to their past
- Qualitative and quantitative data of how this tool has helped those in need

- **Collaboration:**

- Receive updates and offer support and suggestions to those working on the scene
- Tangible advice for making the tool translatable to possible test subjects
- Aid in the coding portion of the scene

# Amanda and Coco's Role

- **Tasks:**

- Document the current scene\_01.json / models.json schema so the team has one clear reference
- Sync with Fahim on the Bennett & Jungu branches before building anything new, to avoid duplicate work
- Researching ways to figure out if there's a faster way of creating models/environments in one prompt /instantaneously through some tech instead of building it from scratch one by one
- Prototype one automation slice: a script that generates a valid scene\_XX.json skeleton from just an avatar list + dialogue script
- Researching ways to drive the scenarios in a way where the users action could “change the storyline”, like a choice based game
- (Semester 2) Turn the script into a lightweight authoring tool so new scenarios don't require hand-editing JSON
- (Semester 2) Extend automation to swap environments, avatars, and dialogue at scale across multiple scenario types

- **Realistic Achievement:**

- A working script that generates a valid scene file from simple inputs — proof that one slice of automation works, not the full pipeline
- A documented, versioned scene-file schema other teammates can build against

- **Collaboration:**

- With Fahim: align on repo structure and branch findings before scoping new tooling
- With Jayden: wire her voice/audio work into the scene timeline's speak and lipSync fields
- With whoever locks the engine choice: confirm whether automation targets VRM/VRMA or the Unreal + Babylon.js pipeline before scaling

# Kevin's Role

- **Tasks:**

- Learning Babylon.js documentation
- Learn Unreal Engine XR and AI system documentation
- Ensuring Babylon and Unreal Engine feature compatibility
- Creating a Blueprint Struct in Unreal to match keys of scene JSON files with JSON Utilities Plugin.
- Integrating project to VR device.
- Working directly with KPC clinicians and clients to understand user needs
- (Next semester) Designing behavior trees for NPCs, automated based on scene JSON
- (Next semester) Designing Smart Objects for scene interactivity.

- **Realistic Achievement:**

- Deployable Babylon VR experience onto Meta Quest 3/3s device.

- **Collaboration:**

- Supporting VR implementation for controllers
- Coco and Amanda: Integrating their JSON file automation to the Blueprint Struct in Unreal Engine.

# Fahim's Role

- **Tasks:**
  - Dissect and refactor GitHub repository for existing features and testing performed in previous semesters.
  - Analyze and correct gap in facial animation and audio synchronization.
  - Coordinate and document team progress, tasks, and collaboration.
  - Ensure accurate compilation and loading of scenes in [Babylon.js](#) using existing builds and characters, noting and adjusting for gaps in performance.
  - Integrate idle animations for character models, potentially using Text to VRMA (<https://github.com/Kirakun0328/text-to-vrma>)
  - Standardize explicit software build process and how data is compiled.
  - Develop validation/testing for scene JSON files, character models, animations, and audio before they are loaded into [Babylon.js](#).
  - Create technical documentation/tutorials for setting up, building, testing, and running the VRET project locally.
- **Realistic Achievement:**
  - Refactored and documented repository structure so future team members can understand the existing pipeline and previous work.
  - Reliable Babylon.js build/demo that loads the existing scene, characters, audio, and animations correctly.
  - Standardized build and deployment workflow with repeatable steps for compiling and testing scenes.
- **Collaboration:**
  - Amanda & Coco: Coordinate repository structure, branches, and scene JSON format so their automation work integrates cleanly with the existing project.
  - Kevin: Work together on integrating scene JSON/Babylon.js work with the Unreal/VR implementation and help troubleshoot compatibility issues.
  - Jayden: Integrate voice files with facial animation/lip-sync and help test synchronization between dialogue and character animation.
  - Whole team: Track GitHub progress, resolve merge/integration problems, document decisions, and compile team's work.

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# How much time should I dedicate to VIP BEAR?

Each student should be dedicating 3 hours per 1 credit of VIP, per University policy. Please indicate in this slide how each team member will dedicate their time.

- **Amanda - VIP 395 1 cred = 3 hours**
  - 1 hr Meeting, 1 hr Independent work for finishing tasks, 1 hr researching ideas
- **Coco - VIP 395 1 cred = 3 hours**
  - 1 hr Meeting, 1 hr Independent work for finishing tasks, 1 hr researching ideas
- **Jayden - VIP 395 1 cred = 3 hours**
  - 1 hr Meeting, 1 hr Independent work for finishing tasks, 1 hr researching ideas
- **Kevin - VIP 395 1 cred = 3 hours**
  - 1 hr Meeting, 1 hr Independent work for finishing tasks, 1 hr researching ideas
- **Fahim - VIP 595/CSE 523 4 cred = 12 hours**
  - 2 hr Meeting, 4 hr Independent work, 3 hr for team coordination, 3 hr researching ideas
- **Carlos - Volunteer = 1~2 hours**
  - 1 hr research, 1 hr work on tasks